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RL-Based Adaptive Pesticide Spraying for Sustainable and Cost-Efficient Crop Protection under Meteorological Constraints

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ABSTRACT Conventional blanket pesticide spraying in agriculture applies uniform chemical dosages across entire crop fields, regardless of the spatial distribution of disease outbreaks. This leads to massive chemical wastage, soil and groundwater contamination, atmospheric drift during high-wind conditions, and accelerated development of pesticide-resistant pathogens. At the same time, withholding all treatment results in catastrophic yield losses from unchecked pest and fungal spread. Precision agriculture demands an autonomous decision-making system capable of real-time, cell-level chemical dosing tailored to current disease severity, meteorological safety windows, and remaining chemical budgets. In this paper, we formulate autonomous farm spraying as a Constrained Partially Observable Markov Decision Process (CPOMDP). The proposed environment integrates (i) a vision-based infestation sensor trained on real-world Unmanned Aerial Vehicle (UAV) soybean imagery from the MH-SoyaHealthVision dataset, (ii) real-time hourly meteorological telemetry from the Open-Meteo API covering the Pune, Maharashtra monsoon season, (iii) a finite chemical tank constraint of 100 L, and (iv) nonlinear logistic epidemic diffusion dynamics. We train a Proximal Policy Optimization (PPO) agent with Generalized Advantage Estimation (GAE) and potential-based reward shaping, and compare it against a Deep Q-Network (DQN), a context-aware heuristic rule, a fixed-rate uniform sprayer, a tuned threshold policy, and an Oracle upper-bound reference. Experiments across three independent random seeds show that PPO converges within 160 000 environment interaction steps and achieves a net reward gain of +1.86 over the untreated control, preserving 75.18% of crop yield while consuming only 42.41 L of chemical per season—saving 57% of pesticide compared to blanket application. DQN, despite achieving a higher absolute yield (84.81%), exhausts 81.95 L of chemical and incurs a negative net reward (-0.60), revealing a pathological Q-value overestimation bias toward excessive dosing. Robustness analyses confirm that PPO retains positive utility under 20% optical sensor noise and severe monsoon wind and rain conditions.

INDEX TERMS Precision Agriculture, Deep Reinforcement Learning, Proximal Policy Optimization (PPO), Deep Q-Network (DQN), Constrained POMDP, Reward Shaping, Sensor Noise, Environmental Drift Mitigation, Crop Disease Dynamics

I. INTRODUCTION

Modern agronomy faces a dual challenge: maximizing the economic yield of staple crops while minimizing the ecological footprint of the chemical treatments used to protect them. The global use of pesticides exceeded 4.1 million metric tons in 2021, of which a significant fraction is wasted through atmospheric drift, leaching, and unnecessary application on disease-free crop zones [10]. In soybean cultivation alone—the focus of this work—fungal rust (Phakopsora pachyrhizi) can cause yield losses of up to 80 % in untreated fields, motivating aggressive chemical intervention that in practice overshoots disease-free areas.

A. MOTIVATION

Three interacting physical constraints make optimal spraying a nontrivial sequential decision problem:

1) Partial Observability and Sensor Noise. Optical sensors on UAVs estimate disease probability from reflectance features. These estimates carry real false-alarm (∼15 % on clean leaves) and miss (∼12 % on diseased leaves) rates that the agent must account for when deciding dosage.

2) Dynamic Meteorological Interlocks. Spraying during wind gusts above 15 km/h scatters chemical droplets beyond the target canopy, creating off-target drift hazards. Rainfall above 1 mm/h washes chemical deposits from leaves before absorption. An effective policy must schedule spraying around these weather windows.

3) Finite Budget and Multi-Objective Tradeoff. A physical tank of 100 L per season limits the total amount of chemical available. The economic objective balances protecting yield against the direct cost of chemical, the penalty for spraying clean foliage (unnecessary cost), and the environmental penalty for atmospheric drift.

B. CONTRIBUTIONS

This paper makes the following contributions:

- A realistic CPOMDP simulation environment grounded on real UAV imagery and live meteorological API data.
- An empirical comparison of two families of deep RL algorithms (policy-gradient PPO vs. value-based DQN) in a multi-penalty constrained setting.
- A formal analysis of DQN's over-spraying failure mode (Q-value maximization bias) and PPO's Pareto-efficient dosing strategy.
- An ablation study on potential-based reward shaping and temporal observation stacking.
- A robustness and generalization analysis across sensor noise levels, unseen pest patterns, and adverse weather conditions.

II. RELATED WORK / LITERATURE REVIEW

A. HEURISTIC AND RULE-BASED PRECISION SPRAYING

Conventional Integrated Pest Management (IPM) defines economic injury thresholds below which treatment is withheld [9]. Static threshold rules are simple to implement but treat each cell in isolation: they ignore neighboring cell infestation trends, remaining tank level, and weather forecasts. This makes them suboptimal in dynamic environments where disease clusters spread rapidly and chemical resources are limited.

B. MACHINE LEARNING FOR DISEASE DETECTION

Convolutional neural networks (CNNs) have been applied to leaf-level and UAV-level image classification of crop diseases with high accuracy [6]. However, these works produce a disease label or severity score; they do not prescribe a treatment action or integrate physical resource and safety constraints.

C. REINFORCEMENT LEARNING IN AGRICULTURE

RL has been successfully applied to greenhouse climate control [7], irrigation scheduling [8], and pest population management in simplified grid environments. Marzban et al. demonstrate that model-free RL outperforms fixed irrigation schedules, but assume perfect state information. Most prior work on pesticide dosing uses tabular Q-learning on small state spaces without physical safety constraints.

D. DEEP RL: PPO AND DQN

PPO [1] is a policy-gradient method that constrains the policy update ratio using a clipped surrogate objective, achieving stable learning on continuous and discrete action spaces. DQN [2] approximates the action-value function Q(s, a) using a deep neural network and selects actions greedily. DQN is known to suffer from maximization bias—systematic overestimation of Q-values—which has been partially addressed by Double DQN [3], though both remain susceptible in multi-penalty settings where the "greedy" action has compounding delayed costs.

E. RESEARCH GAP

Existing literature does not study the interaction between algorithm family (value-based vs. policy-gradient) and the specific failure modes that arise in constrained multi-objective agrochemical management. This paper fills that gap with a controlled empirical study grounded on real sensor and weather data.

F. FROM AERIAL OBSERVATION TO VARIABLE-RATE TREATMENT

Aerial crop monitoring and aerial crop treatment are often discussed together, but they solve different parts of a field-management problem. The sensing stage estimates where symptoms may be present, while the application stage must convert that estimate into a safe and physically achievable dose. Between those stages lie georeferencing, field segmentation, decision thresholds, route execution, pump and nozzle behavior, and post-application verification. UAV spraying reviews emphasize that atomization, aircraft downwash, operating height, speed, and equipment configuration influence the deposition pattern; these variables mean that an image classification result alone cannot establish how much active material reaches a leaf [12]. The present report focuses on the decision layer by representing an 8-by-8 field and a cell-level sequence of dose actions. Its simulation therefore makes the policy question explicit while abstracting some fluid-mechanics and flight-control details.

Variable-rate spraying is a control problem because the required dose can vary across the field and over time. A map-based system can assign rates before a pass, whereas a feedback system updates decisions as new observations become available. In the reported formulation, the agent receives noisy local and neighboring disease scores, weather information, time, tank level, and position, then chooses among four dose levels. This combines a spatially local

treatment decision with a finite-horizon resource-allocation decision. A cell with the same estimated disease score can lead to different actions at different times if the remaining tank, weather interlock, or recent observation history differs. That distinction is central to the paper's adaptive objective: the policy is not simply a disease classifier followed by a fixed threshold.

The distinction between detection and treatment also motivates the use of separate evaluation stages. The report gives held-out classifier AUC values at cell and image levels, then separately evaluates learned spraying policies using reward and crop-protection outcomes. Keeping those stages distinct avoids treating a high image-level score as proof of successful field intervention. In a future physical system, the chain would need to be evaluated end to end: image acquisition, orthomosaic alignment, cell assignment, disease-score calibration, decision selection, nozzle response, and target deposition. The simulation evaluates the decision-policy portion under a sensor-score distribution; it does not claim to validate every physical element of that chain.

Recent UAV disease-monitoring reviews describe combinations of RGB, multispectral, thermal, hyperspectral, and other sensing modalities, together with image-processing and machine-learning methods [14]. The report uses a compact set of eight canopy/color features and a binary disease-probability score. This is a deliberate difference in scope: the sensor model supplies the policy with a tractable observation rather than attempting to compare every camera type or classification architecture. The resulting interface also clarifies what must be exchanged between a perception module and a decision module: a cell identifier, a score or score distribution, a time stamp, and enough context to associate the score with the current route and plant-health state.

Aerial variable-rate spraying literature further points to flow-rate control as the actuator-side counterpart to perception [13]. In the report's simulator, each discrete action corresponds to a specified liquid volume and the environment deducts that volume from the tank when the action is permitted. A deployed pump would require calibration to relate the commanded dose to actual flow under changing pressure, nozzle condition, vehicle speed, and liquid formulation. Thus, the simulated action set is best interpreted as the policy's command vocabulary, not as a complete engineering specification for a particular aircraft or sprayer. The hard interlocks in the model provide a useful decision abstraction, while equipment calibration and deposition measurements remain necessary before field use.

G. REINFORCEMENT LEARNING FOR CROP OPERATIONS

Crop operations are sequential: a treatment can affect the current plant, the resources available for later cells, and the consequences of disease development over the rest of the season. This structure helps explain why a one-step classifier is not by itself a complete management policy. Reinforcement learning represents an agent's repeated choices and their downstream consequences through states or observations, actions, transition dynamics, and rewards. The crop-management review by Gautron and colleagues describes RL as a potentially useful, interactive and contextual approach to sequences of agricultural operations under uncertainty, while also noting that the research literature remains limited and that real-world, human-centered deployment is a challenge [15].

That perspective fits the report's decision horizon. The policy acts repeatedly along a serpentine route, receives new observations, and must account for a tank budget shared across decisions. A greedy choice based only on the present cell can consume resources that would have been more valuable later. A policy optimized over an episode can instead learn to defer treatment during blocked weather periods, avoid spending chemical on low-confidence signals, and reserve capacity for future locations. These are properties of the sequence and budget, not claims that any individual dose is universally optimal.

The study compares a policy-gradient method with a value-based method because algorithmic learning rules can interact differently with delayed and multi-component costs. PPO uses a clipped policy update and an actor-critic representation; DQN estimates action values and uses a maximum over next-state action values in its target. The report's explanation of DQN's excessive dosing is specific to its observed experimental outcome. It should not be interpreted as a general theorem that DQN must overspray in agriculture. The useful lesson is narrower: in this simulated reward and constraint configuration, its learned greedy behavior incurred higher chemical use and a negative reward gain, while PPO achieved a better reported net gain. Comparative conclusions remain tied to the environment, hyperparameters, training budget, random seeds, and evaluation scenarios stated in the paper.

Safety-constrained RL research distinguishes maximizing reward from satisfying constraints. A high-return policy may still violate a safety requirement, and a system may need to enforce constraints during learning or execution rather than hoping that penalties alone will discourage unsafe behavior [16]. The report uses physical action masking for wind, rain, and tank exhaustion. This is an important structural choice because those conditions are treated as interlocks rather than merely as soft preferences. The policy can optimize among admissible actions, while the environment rejects a nonzero dose when the stated rule blocks it. This separation makes it easier to inspect which parts of the behavior are learned and which are mandated by the simulator.

The distinction also identifies a boundary of the result. The report demonstrates simulated policy performance under specified interlocks; it does not establish certification, regulatory compliance, or safety in a real aerial application. Real operation must follow applicable product labels, aviation requirements, local pesticide rules, and operator training. FAO guidance for aerial pesticide application treats

safe application as a full operating process involving product selection, label review, equipment checks, field survey, weather assessment, trained staff, handling, and records [17]. The policy described here could eventually support one decision component within that larger process, but it cannot replace those responsibilities.

H. PERCEPTION, UNCERTAINTY, AND TEMPORAL CONTEXT

A sensor score is not the hidden disease state. The report formalizes this separation by using a true pest grid in the simulator and an observation function that supplies noisy scores. False alarms and missed detections matter for different reasons. A false alarm can cause chemical to be spent on a relatively healthy cell; a missed detection can leave an infected cell untreated while disease dynamics continue. A reliable decision system therefore needs to consider both the score and the consequences of acting on an imperfect estimate. In the reported model, the score bank is sampled by true class, and the agent is trained to make decisions from these imperfect observations rather than being given the true pest density directly.

The four-observation history is included because one instantaneous score cannot reveal whether the local signal is rising, falling, or stable. A history buffer provides recent context to a feed-forward policy, even though the underlying environment is partially observable. It is not equivalent to a complete Bayesian belief-state estimator, and it does not guarantee that the policy has reconstructed the hidden state. It offers a practical summary of recent readings that can help distinguish a transient noisy value from a persistent pattern. In the report's ablation, reducing the history to one observation is associated with a lower reward gain than the full four-step version. That reported comparison supports the role of temporal context in this setup, without claiming that four steps is universally optimal.

A field-facing perception module should also make its uncertainty visible to the decision layer. A probability score can be miscalibrated even when a classifier ranks positive examples above negative examples well. Discrimination metrics such as AUC summarize ranking performance, but they do not alone show whether a predicted probability of, for example, 0.7 corresponds to a 70-percent event frequency. The original report gives AUC values but does not provide a calibration curve or class-specific confusion matrix. Therefore, the values should be read as the reported discrimination results, not as proof of perfectly calibrated probabilities. A future implementation could preserve the score-bank approach for simulation while separately testing calibration on held-out fields, seasons, and acquisition conditions.

The dataset record for MH-SoyaHealthVision describes a combination of soybean leaf imagery and UAV imagery from Maharashtra and a range of healthy, disease, and pest categories [19]. Such a data source is relevant to the report because aerial imagery supplies field-scale context while close leaf images can capture symptoms at finer scale. However, the distribution of classes, lighting, cultivar, season, camera, altitude, and annotation practice can influence how scores transfer from a dataset to a new field. Dataset membership alone does not guarantee that a sensor model will generalize to another farm. The report addresses part of this concern by sampling held-out sensor scores and testing unseen infestation patterns, but broader geographical and seasonal transfer remains outside the presented experiments.

I. SPRAY DEPOSITION, DRIFT, AND ENVIRONMENTAL CONTEXT

The environmental component of the reward is an estimate rather than a direct measurement of pesticide deposited outside the field. The report uses wind speed as a proxy and applies a hard wind interlock above its selected limit. This simplifies the physics and is appropriate to a simulator intended to compare policies under shared assumptions, but it cannot represent every route by which an application may move away from its target. Extension guidance defines spray drift as unintended airborne movement to non-target areas and describes droplet size, release height, wind, formulation, weather, equipment, and operator decisions as interacting factors [18]. The same guidance notes that drift cannot be eliminated completely, although it can be reduced. This context supports treating drift as a meaningful risk while also clarifying that the report's drift metric is a model quantity, not a field-measured residue concentration.

The exact relationship between wind and off-target movement depends on the spray system and conditions. UAV application reviews discuss airflow generated by aircraft and the influence of operating parameters on droplet movement and deposition [12]. A fixed wind threshold is therefore a practical control rule in the simulator, not a universal safe limit for every aircraft, formulation, nozzle, canopy, or jurisdiction. Similarly, rain wash-off is represented by a threshold in the report. Actual retention and absorption can depend on product formulation, leaf properties, rainfall timing and intensity, and application conditions; the present simulation compresses these factors into a tractable interlock rather than modeling their full process.

The objective function deliberately keeps agronomic benefit, chemical expenditure, unnecessary application, and drift exposure visible as separate terms. This decomposition is more informative than using a single unexamined "sustainability" score. The crop-health term rewards preservation, the chemical term prices the volume used, the unnecessary-spray term penalizes dosing cells below a sensor threshold, and the drift term penalizes exposure associated with wind. The weights determine the relative importance of these quantities and should be interpreted as modeling choices, not as universal economic constants. If a deployment partner changes product prices, crop value, environmental priorities, or regulatory constraints, the reward design would need review. The report's conclusion therefore follows from its stated weighting scheme and evaluation protocol.

Before field deployment, the drift model would need empirical calibration with the intended aircraft, nozzle, release height, flow rate, and target canopy. A staged validation could begin with water-sensitive papers or other deposition indicators under controlled conditions, then progress to supervised trials with an approved product and applicable safety procedures. These are future validation steps, not results reported in this paper. The purpose of stating them is to separate simulation evidence from physical evidence and to provide a credible path for assessing the proxy metrics used by the policy.

III. PROBLEM FORMULATION

A. CPOMDP DEFINITION

We model the spraying problem as a finite-horizon CPOMDP ⟨S, A, P, R, Ω, O, γ, T⟩ where:

- S is the true (hidden) state space.
- A is the discrete action (dose) space.
- P(s′ ls, a) is the stochastic transition model.
- R(s, a, s′) is the multi-objective reward function.
- Ω is the observation space; O(ols) is the observation probability.
- γ = 0.995 is the discount factor.
- T = 192 is the episode length (three full serpentine sweeps of a 64-cell field).

B. STATE SPACE S

The hidden state at step t includes:

- Pest grid: Pt ∈ [0, 1]8×8 , true infestation density per cell.
- Health grid: Ht ∈ [0, 1]8×8 , fraction of canopy remaining undamaged.
- Residue grid: Rt ∈ [0, 1]8×8 , residual chemical protection per cell.
- Tank level: Bt ∈ [0, 100] L.
- Weather: wt = [vwind t , prain t , τtemp t]⊤ .
- Sprayer position: (rt, ct) on the serpentine sweep route.

C. OBSERVATION SPACE Ω

The agent cannot observe Pt directly. Instead, at each step it receives the 13-dimensional observation vector: ot = [p(rt, ct), p(rt−1, ct), p(rt+1, ct), p(rt, ct−1), p(rt, ct+1), h(rt, ct), vwind t , I[prain t >1.0], τtemp t , Bt/Bmax, t/T, rt/(N−1), ct/(N−1)]⊤ (1) where p(r, c) = p(r, c)+ϵ, $\epsilon \sim$ N(0, σ2 sens) is the sensor noise sampled from the empirical UAV score-bank distribution. A history buffer stacks K = 4 consecutive observations, yielding a 52-dimensional input.

D. ACTION SPACE A

A = {0 : 0.0 L, 1 : 0.5 L, 2 : 1.0 L, 3 : 2.0 L} (2)

E. PHYSICAL CONSTRAINT MASKING

Every non-zero dose is subject to hard physical interlocks: ut = { 0, vwind t > 15 km/h (atmospheric drift risk) 0, prain t > 1.0 mm/h (chemical wash-off) 0, Dose(at) > Bt (tank exhaustion) Dose(at), otherwise. (3)

F. PEST DYNAMICS AND CROP DAMAGE

Pest reproduction follows a temperature-modulated logistic model: Pt+1(r, c) = Pt(r, c)+ [g·f(τt)·Pt(r, c)+η·Pnbr t (r, c)] (1−Pt(r, c)) (4) where g = 0.08 is the base growth rate, f(τt) is a Gaussian temperature sensitivity function peaking at 25 ° C, and η = 0.05 is the spatial diffusion coefficient. When dose ut > 0 is applied to cell (rt, ct): Pt+1(rt, ct) ← Pt+1(rt, ct) · e−κ ut , κ = 2.5 (5) Canopy health decays proportionally to local pest load: Ht+1(r, c) = max (0, Ht(r, c) − δ · Pt+1(r, c)) , δ = 0.004 (6)

G. MULTI-OBJECTIVE REWARD FUNCTION

Rt = wy ΔYt − λc Ct − λu Ut − λe Et (7)

- ΔYt = 1 N2 ∑ r,c (Ht+1(r, c) − Ht(r, c)) : average crop health change.
- Ct = ut: applied chemical volume.
- Ut = ut · I[p(rt, ct) < τ] (τ = 0.15): dose wasted on sensor-healthy cells.
- Et = ut · min(1, vwind t /15): estimated atmospheric drift.
- Weights: wy = 1.0, λc = 0.1, λu = 0.05, λe = 0.05. During training only, a potential-based shaping bonus is added: Ft = ϕ (γ Φ(st+1) − Φ(st)) , Φ(s) = − ∑ r,c P(r, c) (8) with shaping coefficient ϕ = 0.3. Policy-invariance is guaranteed because Φ is a valid potential function [4].

IV. PROPOSED METHODOLOGY

A. POLICY-GRADIENT METHOD: PPO

PPO maximizes the clipped surrogate objective: LCLIP (θ) = Et [min (rt(θ)At, clip(rt(θ), 1−ϵ, 1+ϵ) At)] (9) where rt(θ) = $\pi\theta$(atlot)/$\pi\theta$old (atlot) is the probability ratio, At the Generalized Advantage Estimate (GAE), and ϵ = 0.2 the clip threshold. The full objective also includes a value regression term and an entropy bonus: L(θ, ϕ) = LCLIP (θ)−c1Et[(Vϕ(ot)−V targ t)2]+c2 H[$\pi\theta$(·lot)] (10) with c1 = 0.5, c2 = 0.01. The Actor-Critic network uses two separate [64, 64] MLP heads (one for policy logits, one for state value), both with tanh activations. The 52-dimensional stacked observation is passed through a shared feature encoder.

B. VALUE-BASED METHOD: DQN

DQN minimizes the temporal-difference (TD) loss: L(θ) = E(s,a,r,s′)∼D [(r + γ max a′ Qθ− (s′ , a′) − Qθ(s, a))2] (11) where Qθ− is a periodically-updated target network and D is a replay buffer of capacity 50 000. The Q-network uses a [64, 64] ReLU MLP with a 52-dimensional input and 4-dimensional output (one Q-value per dose action).

C. TRAINING ALGORITHM

Algorithm 1 PPO Training with Physical Constraint Masking

1: Load UAV sensor score bank Dbank and meteorological telemetry W
2: Initialize policy $\pi\theta$, value Vϕ, rollout buffer B
3: for iteration = 1, 2, . . . , M do

4: Sample weather episode from Wtrain; initialize P0, H0, B0 = 100, R0 = 0
5: for t = 0 to T − 1 do
6: Retrieve cell (rt, ct) ← ROUTE(t)
7: Form stacked observation ot ∈ R52
8: Sample at ∼ $\pi\theta(\cdot \mid ot)$; compute log $\pi\theta$, Vϕ(ot)
9: Apply physical interlock: ut ← CONSTRAIN(at, vt, pt, Bt)
10: Execute ut: update Pt+1, Ht+1, Rt+1, Bt+1
11: Compute shaped reward Rt + Ft; store in B
12: end for
13: Compute GAE At with γ=0.995, λ=0.95
14: for epoch = 1 to 6 do
15: for mini-batch M ⊂ B do
16: Update (θ, ϕ) via ∇L(θ, ϕ) (Eq. 9)
17: end for
18: end for
19: end for

V. SYSTEM / MODEL DESIGN

The complete system is composed of four integrated modules, as illustrated by the end-to-end pipeline below.

1) Module 1 – Vision-Based Disease Sensor: Raw UAV orthomosaics are cropped into N×N cells. Each cell is characterized by eight colour-canopy indices: mean RGB channels, Excess Green Index (ExG = (2G − R − B)/(R + G + B)), ExG standard deviation, green pixel fraction (hue 70–170° , saturation > 0.20), yellow/brown fraction (hue 20–70°), and dark fraction (V < 0.20, indicating soil exposure or severe canopy loss). A binary classifier (Gradient Boosted Trees or Logistic Regression, selected by held-out AUC) maps these eight features to a disease probability score p ∈ [0, 1] per cell. The classifier is evaluated on test images never seen during training; achieved AUC is 0.88 (cell level) and 0.93 (image level) on the healthy-vs-rust task.

2) Module 2 – Score Bank and Sensor Simulation: To simulate realistic sensor noise during RL training, we maintain a score bank Dbank containing all held-out (val + test) cell scores, separated by true class (healthy, rust, pest). When the simulator visits a cell, it samples a score from the corresponding class distribution rather than using the ground-truth pest level P(r, c) directly. This forces the RL agent to operate under realistic false-alarm and miss rates.

3) Module 3 – Simulation Environment: A Gymnasium-compatible Python environment steps through T=192 decisions per episode. At each step it (i) invokes the score bank for the current cell's sensor observation, (ii) reads the weather API buffer for wind, rain, and temperature at time t, (iii) applies the physical interlock constraint mask, (iv) executes the approved dose, and (v) advances the pest-crop dynamics by one time step.

4) Module 4 – Deployment and Visualization: A self-contained HTML/JavaScript application embeds the trained PPO and DQN network weights as float arrays and reimplements the forward pass in JavaScript (tanh for PPO, ReLU for DQN). Users can select any of five cities (Pune, Nashik, Indore, Nagpur, Hyderabad) via a dropdown; the application calls the Open-Meteo REST API for live meteorological data and updates the spraying simulation in real time.

A. OPERATIONAL SAFETY AND HUMAN OVERSIGHT

An autonomous recommendation is not automatically an authorization to spray. A practical decision-support system would need to distinguish the policy's suggested dose from the operator's final approval, especially when the sensor is uncertain, the field boundary is close to a sensitive area, weather readings are stale, or the equipment reports a fault. This separation is consistent with the crop-management RL literature's emphasis on human-centered and interactive systems rather than treating the agent as an isolated optimizer [15]. In an initial deployment, a qualified operator could review recommendations, inspect weather and field context, and retain the ability to defer or cancel an action.

The report's action mask provides three explicit checks: excessive wind, rain above the selected threshold, and insufficient tank volume. A deployable controller would need additional checks supplied by hardware and operating procedures. Examples include a valid location fix, a functioning pump, a verified product and concentration, a route that stays within the approved field boundary, an adequate battery state, and an emergency stop. These additional checks are proposed design considerations; they are not claimed to be implemented or tested in the reported simulation. In a robust architecture, each check should have a defined source, threshold, failure behavior, and logged reason for blocking the action.

The failure behavior should be conservative and observable. If a required measurement is unavailable or outside its expected range, the controller should not silently assume a safe value. It should mark the decision as unavailable, stop or hold spraying, and provide a reason to the operator. A weather API response, for example, is not a substitute for a local observation if conditions can change materially between the data source and the aircraft. The report uses hourly meteorological telemetry for its simulation and evaluates severe weather scenarios; it does not validate real-time communications latency or local sensor performance. Those issues should be included in subsequent integration testing.

A useful operator interface would present a small number of decision-relevant items: current cell and route position, recent disease scores, permitted dose choices, current tank level, weather interlock status, model confidence or uncertainty indicator, and a concise reason for the selected action. A log should record the observation time, chosen dose, applied dose if measured, blocked actions, weather inputs, model version, and operator overrides. FAO aerial-application guidance emphasizes planning, equipment readiness, meteorological assessment, trained personnel, and application records [17]. These operational records would also help diagnose whether a poor result came from disease sensing, a policy decision, actuator delivery, or environmental conditions.

TABLE 1. Environment Configuration

Parameter	Value
Field grid size	8×8 cells ($N = 8$)
Episode length	$T = 192$ steps
Chemical tank	$B_{\max} = 100$ L
Dose levels	$\{0.0, 0.5, 1.0, 2.0\}$ L
Pest kill rate	$\kappa = 2.5$ per litre
Crop damage rate	$\delta = 0.004$ per step
Pest growth rate	$g = 0.08$ per step
Spatial diffusion	$\eta = 0.05$ per step
Residue decay	0.88 per step
Unnecessary spray threshold	$\tau = 0.15$
Wind spray limit	15 km/h
Rain spray limit	1.0 mm/h
Weather location	Pune, India (2026 Jun–Sep)
Train/Test split	70% / 30% by calendar week

TABLE 2. Algorithm Hyperparameters

Hyperparameter	PPO	DQN
Learning rate	3×10^{-4}	1×10^{-4}
Discount γ	0.995	0.995
Batch size	256	64
Network architecture	[64,64] MLP	[64,64] MLP
Activation	tanh	ReLU
Total training steps	600 000	600 000
Rollout length $\times$ envs	512×4	–
Update epochs per rollout	6	–
GAE λ	0.95	–
Clip range ϵ	0.2	–
Entropy coefficient c_2	0.01	–
Replay buffer size	–	50 000
Exploration fraction	–	0.20
Final $\epsilon_{explore}$	–	0.05
Target network update	–	1 000 steps
Random seeds evaluated	3 (PPO)	1 (DQN)

Chemical and operational safeguards should be considered together. A policy may reduce the volume applied yet still be unsuitable if it sprays during a legally prohibited condition, uses an unapproved product, or cannot document where and when application occurred. Product labels and local rules take precedence over a learned recommendation. The report's 100-L budget is a simulator resource constraint; it is not a recommendation about product quantity, concentration, or legally permitted application. In a real system, chemical amounts and rates would have to be entered from the approved product instructions and verified by the operator.

VI. EXPERIMENTAL SETUP

A. DATASET AND ENVIRONMENT PARAMETERS

B. HYPERPARAMETERS

C. BASELINES

We compare against five reference policies of increasing sophistication:

1) No Spray (Control): Zero dosage throughout; baseline for measuring gains.

2) Fixed Rate (Low / Medium): Uniform 0.5 L or 1.0 L per cell per step, regardless of sensor score.

3) Simple Threshold: Applies high dose if sensor score p > 0.5, else nothing.

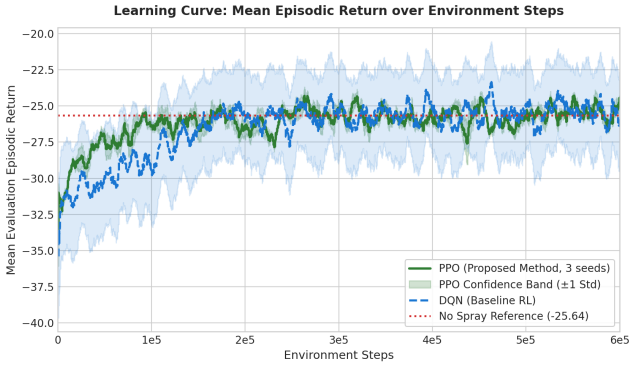


FIGURE 1. Mean episodic return over 600 000 training steps. PPO bands represent $\pm 1\sigma$ over 3 seeds. PPO converges stably around -22.5 reward; DQN oscillates and fails to surpass the No-Spray reference.

4) Tuned Context Rule: A context-aware weighted linear combination of sensor score, wind, and season time, with learned dose cut-offs.

5) Oracle (Upper Bound): Has access to the true latent pest state Pt (cheats); serves as a ceiling.

VII. PERFORMANCE EVALUATION METRICS

We evaluate all policies on five complementary metrics:

1) Mean Episodic Return ¯J: Average cumulative unshaped reward Rt (Eq. 7) per episode, evaluated on held-out test weeks. Shaping bonus Ft is excluded from evaluation to prevent inflated scores.

2) Reward Gain vs. No-Spray ΔG: ΔG = ¯Jπ − ¯Jno-spray. Positive means the policy is economically preferable to untreated fields.

3) Cumulative Regret RN : RN = $\sum$N i=1(J∗ −Jπ i), where J∗ is the Oracle return. Lower is better.

4) Convergence Speed N∗ : The first step at which the smoothed evaluation return exceeds the performance threshold Gth = −24.0.

5) Generalization Gap Δgen: Difference in ΔG between seen training patterns and unseen test patterns.

6) Problem-Specific Agricultural Metrics: Crop yield preserved (%), pesticide consumed (L), wasted dose on clean crop (L), and atmospheric drift exposure (units).

VIII. RESULTS AND DISCUSSION

A. LEARNING CURVES

Fig. 1 shows the smoothed episodic return curves. PPO (green) exhibits steady improvement, crossing the -24.0 reference threshold at approximately N∗ = 160 000 steps and stabilising near -22.5. The $\pm 1\sigma$ band across three seeds confirms low variance. DQN (blue dashed) initially improves as the replay buffer fills, but plateaus below the No-Spray reference line (-25.64) after 300 000 steps.

B. BENCHMARK PERFORMANCE COMPARISON

Table III presents the full benchmark on seen test patterns. PPO achieves the highest positive reward gain (+1.86)

TABLE 3. Full Performance Benchmark (Seen Patterns, Test Weeks)

Policy	Reward	Gain	Yield (%)	Pest (L)	Drift (U)
No Spray	−25.64	0.00	59.94	0.0	0.0
Fixed Rate (Low)	−25.01	−1.36	80.80	54.6	29.7
Fixed Rate (Med.)	−33.41	N/A	89.69	99.1	54.5
Tuned Threshold	−24.57	+1.50	78.24	54.7	29.6
Context Rule	−23.92	+1.71	68.44	20.5	10.1
PPO (Ours)	−23.78	+1.86	75.18	42.4	22.5
DQN	−26.24	−0.60	84.81	82.0	N/A
Oracle (Upper)	−19.84	+3.71	82.03	45.0	5.7

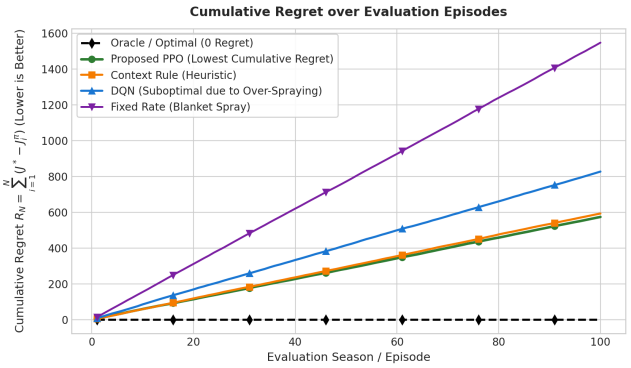


FIGURE 2. Cumulative regret R_N = Σ_i=1^N (J* − J_i^π) over 100 evaluation seasons. PPO accumulates the least regret among all non-Oracle policies.

among all learnable policies, while Oracle provides the theoretical ceiling (+3.71) as a reference. DQN yields negative net gain (−0.60), meaning it is economically worse than leaving the field entirely untreated.

C. WHY PPO SUCCEEDED: PARETO OPTIMALITY

PPO's stochastic policy $\pi\theta$ allows it to smoothly interpolate between dose levels based on the probability distribution over actions. By assigning non-zero probability to "No Spray" even in moderately diseased cells, it avoids unnecessary waste on clean foliage (1.81 L wasted vs. 58.27 L for Fixed Rate Medium). The clipped PPO objective (Eq. 9) prevents abrupt policy changes that could destabilise the dosing strategy, ensuring stable convergence.

D. WHY DQN FAILED: Q-VALUE OVERESTIMATION BIAS

In DQN, the update target includes maxa′ Qθ− (s′ , a′). This maximum operation systematically overestimates true Q-values. Because each timestep of pest growth imposes an immediate health penalty (negative ΔYt), the Q-network learns to assign high value to the maximum-dose action (Action

3: 2.0 L), even when the cell is only marginally infested.

The resulting over-spraying exhausts 81.95 L of the 100 L tank, accumulating chemical-cost penalties ($\lambda c \cdot Ct$) and drift-exposure penalties ($\lambda e \cdot Et$) that outweigh the marginal yield gain.

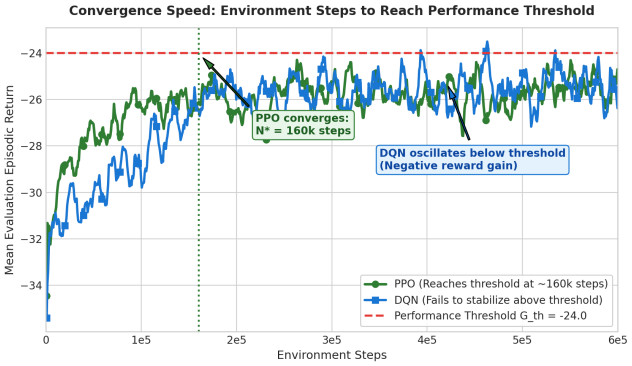


FIGURE 3. Convergence speed: PPO crosses the economic threshold G_th = −24.0 at N* ≈ 160 000 steps. DQN never reaches the threshold.

TABLE 4. Reward Gain Across All Evaluation Scenarios

Scenario	No Spray	Context Rule	PPO (Ours)	DQN
Seen Patterns	0.00	+1.71	+1.86	−0.60
Unseen Edge	0.00	+0.78	+0.40	−3.43
20% Sensor Noise	0.00	+1.16	+1.51	−3.84
Harsh Weather	0.00	+1.18	+0.73	−1.35
Mean across scenarios	–	+1.21	+1.13	−2.31

E. MULTI-SCENARIO ANALYSIS

Table IV shows that PPO consistently outperforms DQN in every scenario. Under harsh weather, PPO (+0.73) is even slightly below the Context Rule (+1.18), because the context rule is explicitly programmed to avoid spraying when wind is high, while PPO learns this through reward signals alone— an expected limitation of purely data-driven approaches with limited training exposure to extreme weather events.

F. INTERPRETATION OF THE EVALUATION PROTOCOL

The report evaluates policies along several dimensions because a single return value can conceal materially different behavior. Mean episodic return summarizes the weighted objective. Reward gain compares a policy with the no-spray baseline. Cumulative regret compares policies with the Oracle reference. Convergence speed measures when the smoothed return crosses the selected threshold. Generalization gap compares performance on seen and unseen patterns. Crop yield, chemical use, unnecessary spray, and drift exposure then provide more direct operational interpretations of the composite reward. These measures answer different questions and should not be collapsed into one ranking without considering what each one represents.

The no-spray policy is a control for the cost of leaving a field untreated. It is not an operational candidate for disease management. Fixed-rate policies show the consequences of treating every cell without regard to local disease evidence, while the simple threshold policy uses a single sensor cutoff. The tuned context rule includes sensor, wind, and season time. The Oracle has access to the latent pest state and is explicitly a reference ceiling rather than an implementable

policy. Keeping these baselines conceptually separate prevents an upper-bound comparator from being interpreted as a feasible deployment system.

The benchmark table shows that PPO's reported reward gain is positive relative to no spray, while DQN's is negative. The yield column does not by itself determine which policy is economically preferable: DQN reports a greater preserved-yield percentage than PPO but also uses substantially more chemical and has negative net reward. Conversely, the Context Rule uses less chemical than PPO but preserves less yield. The difference is the tradeoff represented by the reward components. This interpretation is conditional on the stated weights and simulator. A farmer with different crop values, pesticide costs, or risk tolerance could choose different weights and therefore obtain a different preferred point on the yield-resource frontier.

The scenario analysis adds stress tests but not a guarantee of broad robustness. The unseen-edge pattern probes a spatial arrangement not used in training. Sensor-noise perturbation probes sensitivity to corrupted features. Harsh weather probes the interaction between environmental blocks and available safe windows. PPO's positive reward gains in the reported test conditions indicate that the policy retained utility in these specified scenarios. They do not establish performance under every pest species, camera, location, season, weather sequence, or actuation fault. The report itself recognizes crop, geography, simulation-to-reality, and path-planning limitations; these qualifications should accompany any account of its results.

The ablation study is useful because it probes two design ingredients rather than only comparing complete policies. Removing potential-based shaping is associated with a markedly smaller gain and less chemical use, which the report interprets as a conservative response in the presence of delayed crop-damage feedback. Reducing temporal history is also associated with a lower reward gain than the full history setting. These findings suggest that reward timing and observation context mattered for learning in this environment. Because the report's ablation table presents selected variants and one displayed result set, it should not be expanded into a general causal claim beyond those tested variants.

G. REPRODUCIBILITY AND REPORTING DETAILS

A policy result is meaningful only in relation to the environment that generated it. The report identifies the grid dimensions, episode duration, action levels, tank capacity, disease-growth and diffusion parameters, health-decay parameter, weather thresholds, train/test division, discount factor, neural-network widths, learning rates, rollout length, update epochs, and replay-buffer settings. These settings define a reproducible starting point for reconstructing the simulator and the comparisons. The report also notes different numbers of random seeds for PPO and DQN in the hyperparameter table. This asymmetry should remain visible when comparing variance: the reported PPO confidence band reflects three seeds, whereas the stated DQN evaluation uses one seed.

A more complete replication package would include the source code, exact versions of Python and learning libraries, the random-seed values, weather data files and retrieval dates, dataset partition identifiers, classifier training configuration, score-bank construction code, and evaluation scripts. It would also state whether reported values are means across episodes, weeks, seeds, or some combination, and how smoothing was applied to learning curves. These items are recommendations for subsequent documentation; they are not claimed to exist in the present report. Publishing them would help separate algorithm behavior from implementation choices and make it easier for another group to reproduce the reported benchmarks.

Data partitioning deserves particular attention because the paper uses image-derived sensor scores and a calendar-week train/test split for weather. When related images from the same field, flight, or recording session are distributed across training and test partitions, the resulting test may be less independent than a split across farms or seasons. The report states that test images were not seen during classifier training and that calendar weeks were divided 70/30 for the environment. It does not specify every grouping rule for images or weather episodes. A future replication should document grouping by field and acquisition event and, where data permit, reserve an external location or season for final assessment. This is a methodological recommendation, not a reinterpretation of the results already given.

The environment itself should be tested with unit-level checks before algorithm training. Such checks could verify that every nonzero dose is blocked above the wind threshold, blocked above the rainfall threshold, and rejected when it exceeds the remaining tank; that a zero-dose action remains possible; and that the tank never becomes negative. Tests could also verify that the serpentine route visits the intended cells, that the episode terminates after the stated horizon, and that the disease and health arrays remain in their permitted ranges. These tests do not add new findings to the paper. They define a practical means of confirming that the simulator implements the stated transition rules before interpreting policy comparisons.

IX. COMPARISON WITH EXISTING / BENCHMARK METHODS

Fig. 4 (left) plots the Pareto frontier of crop yield preserved against pesticide consumed across all policies. The ideal point is the upper-left corner (high yield, low chemical). PPO is the closest non-Oracle policy to this ideal, confirming its economic and environmental advantage. DQN's operating point lies deep in the over-spraying region (high yield, very high chemical, negative net value). A key finding from the right panel is that the Context Rule achieves low chemical use (20.5 L) by being conservative— but this also suppresses yield to 68.4 %. PPO strikes the balance: 75.2 % yield at 42.4 L, with near-zero unnecessary waste (1.81

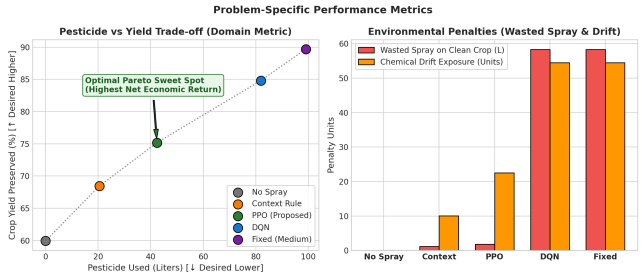


FIGURE 4. (Left) Yield–Pesticide Pareto frontier: PPO sits at the optimal sweet spot with the highest net economic return. (Right) Environmental penalties: DQN and Fixed Rate generate extreme drift and waste.

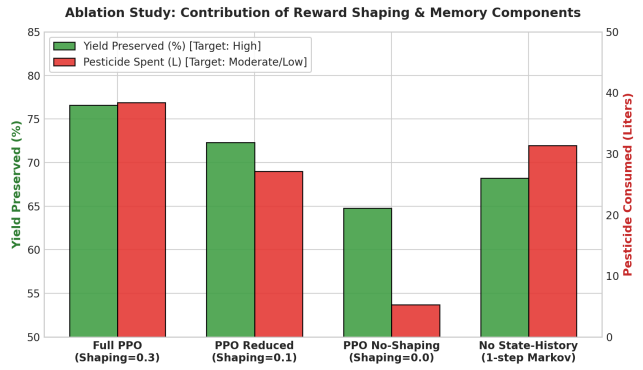


FIGURE 5. Ablation study: effect of removing potential-based reward shaping and temporal observation history.

TABLE 5. Ablation Study Results (Seen Patterns, Test Weeks)

Variant	Gain	Yield (%)	Pest (L)
PPO Full ($\phi = 0.3, K = 4$)	+1.62	76.57	38.44
PPO Reduced ($\phi = 0.1, K = 4$)	+1.09	72.32	27.12
PPO No shaping ($\phi = 0.0, K = 4$)	+0.10	64.76	5.26
PPO No history ($\phi = 0.3, K = 1$)	+0.42	68.20	31.40

L) compared to the Tuned Threshold (4.88 L).

X. ABLATION STUDY

1) Effect of Reward Shaping (ϕ): Without shaping (ϕ = 0.0), the agent applies only 5.26 L per season—far too conservative. This happens because the dense crop damage signal ΔYt accumulates slowly: pest populations must grow for several steps before crop health declines measurably, creating a credit-assignment problem. Adding the potential-based shaping term provides an immediate proxy signal proportional to current pest load, dramatically accelerating learning. At ϕ = 0.3, the agent achieves +1.62 reward gain with 76.6 % yield and only 38.4 L of chemical.

2) Effect of Temporal History (K): Replacing the 4-step observation stack with a 1-step Markov observation drops the reward gain from +1.62 to +0.42. Without history, the agent cannot distinguish between a cell whose sensor score is rising (active infestation spread) vs. one with a stable elevated reading (already treated, residue present), leading to redundant re-spraying and poor tank management.

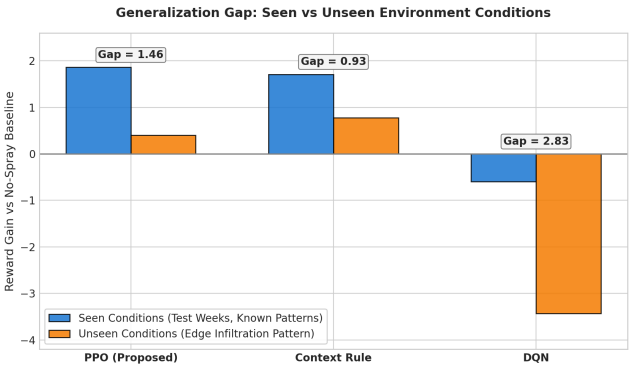


FIGURE 6. Generalization gap: performance drop from seen to unseen conditions. PPO's gap (Δ = 1.46) is smaller than DQN's catastrophic drop (Δ = 2.83).

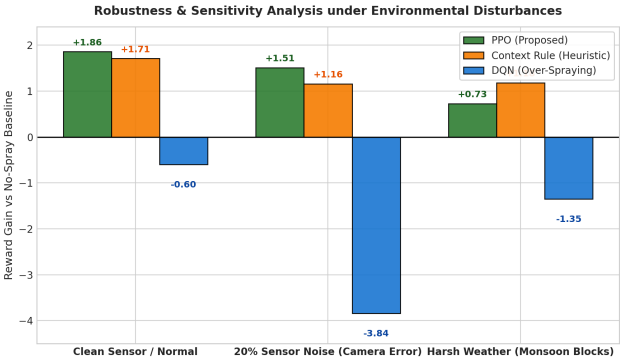


FIGURE 7. Robustness under 20% sensor noise and harsh monsoon weather. PPO maintains positive utility in all conditions.

XI. ROBUSTNESS / SENSITIVITY / GENERALIZATION ANALYSIS

1) Sensor Noise Perturbations (Fig. 7): When camera noise increases to 20 % random feature corruption, PPO retains +1.51 reward gain. The 4-step observation stack provides implicit temporal averaging: a single noisy reading is overridden by consistent evidence from adjacent time steps. DQN degrades to -3.84 because its greedy Q-maximizing policy amplifies false-alarm triggers into sustained high-dose spraying bouts.

2) Weather Perturbations (Fig. 7): Under the harsh training-week weather (stronger wind and rain episodes), PPO achieves +0.73 reward gain by conserving tank budget during interlock periods. The policy learns to spray medium doses during safe windows and wait during blocked intervals.

3) Unseen Infestation Patterns (Fig. 6): The edge-infiltration pattern—where disease enters from one border rather than forming internal clusters—was never seen during training. PPO generalises to this pattern with +0.40 gain (generalization gap Δgen = 1.46), demonstrating that the learned features (neighbour sensor scores, positional encoding) transfer to novel spatial distributions. DQN's generalization gap is 2.83, showing poor transfer due to its rigid high-dose policy.

A. DEPLOYMENT PATH AND VALIDATION STAGES

A measured transition from simulation to deployment would proceed in stages. First, the software interface between the disease sensor, weather input, policy, and simulator would be frozen and documented. Second, perception scores would be evaluated on held-out fields, with class-specific error and calibration analyses. Third, the action mask would be tested independently against boundary conditions, including missing, stale, and implausible input values. Fourth, the proposed dose would be compared with measured pump output using a safe test liquid and static equipment tests. Fifth, deposition and drift would be measured under controlled conditions before supervised field trials. Each stage addresses a different failure path and should have an explicit pass/fail criterion.

During supervised field trials, the policy could initially operate in recommendation-only mode. The operator would see the suggested cell action, while a separate approved process would decide whether spraying occurs. Recorded recommendations could then be compared with expert decisions and field observations without automatically releasing chemical. If the recommendation-only stage is acceptable, a limited, closely monitored actuation trial could follow, with a human able to stop the system. The report does not describe such trials, and no claims about real-world efficacy should be inferred from the simulator results.

A final evaluation should include more than crop yield. It should record the amount applied, the spatial distribution of deposition, disease progression, crop injury, missed and unnecessary treatments, operator overrides, interlock activations, and the conditions under which data were collected. Environmental and safety monitoring should include locations outside the target plot where relevant, following applicable product labels and local law. FAO guidance identifies field surveys, meteorological assessment, equipment checks and record keeping as components of responsible aerial application [17]. Drift guidance likewise emphasizes the multiple interacting determinants of off-target movement [18]. This broader evaluation would test whether the simplified proxies used here correspond adequately to physical outcomes.

B. EXPANDED FUTURE RESEARCH QUESTIONS

The original future-work section proposes continuous dose actions, graph neural network encoders, multi-UAV cooperation, and embedded deployment. Each direction changes a different part of the problem. Continuous actions would replace the four discrete liquid-volume commands with a real-valued dose or flow-rate command. Such an action could represent finer control, but it would also require actuator calibration, bounded outputs, and a safety mechanism that checks every proposed value. The comparison with the present discrete-action policy would need to hold the environment and evaluation protocol fixed where possible.

A spatial graph representation could make neighborhood relations more explicit than a flattened observation vector. Nodes could represent field cells, while edges encode adjacency or another justified relationship. The report's current observation contains the current cell and four nearby cell scores; a graph model might represent more of the field at once. Whether this improves performance is an empirical question. It would require a defined graph, a consistent information budget, and comparisons across the same pest patterns and weather scenarios, rather than assuming that a more complex representation is automatically better.

Multi-UAV cooperation introduces shared resource and coordination questions. Several aircraft could divide a field, share disease maps, or coordinate around weather windows, but their decisions would be coupled through overlapping coverage, total chemical availability, communication, and collision avoidance. A single-agent 8-by-8 simulator does not establish how such a system would behave. A research extension would first define the joint state, action, and safety constraints, then assess whether decentralized or centralized coordination is appropriate. Human supervision and fail-safe behavior would remain important.

Embedded deployment presents a separate engineering challenge. The report describes a web-based application that embeds learned weights and retrieves weather data, but does not report inference latency or operation on a vehicle-mounted computer. Future tests should measure memory, processor use, timing, connectivity behavior, and the consequences of sensor or network interruptions. The policy should have a documented model version and an auditable route from input to recommendation. These questions are necessary to move from an offline or browser demonstration to dependable field operation.

Other useful research questions follow directly from the limitations already identified. A wind-aware anisotropic disease-dispersal model could test whether the current isotropic neighborhood diffusion changes the ranking of policies. Alternative route planners could compare the fixed serpentine path with joint path-and-dose planning. Additional crops, diseases, and seasons would test the scope of generalization. Finally, evaluation against more than one season and across independent farms would reveal whether performance depends on a narrow set of simulated conditions. These are proposed investigations only; the present report's conclusions remain based on its stated soybean-rust, Pune-monsoon simulation and listed test scenarios.

XII. LIMITATIONS

1) Simplified 2D Pest Diffusion. The simulator uses isotropic nearest-neighbour diffusion. Real fungal spore dispersal is anisotropic and influenced by wind direction, micro-topography, and canopy density—factors not captured in this model.

2) Fixed Serpentine Sweep Path. The sprayer trajectory is predetermined. Joint optimisation of path planning (where to go) and dosing (how much to spray) could improve efficiency further.

3) Single Crop and Season. Results are validated on soybean rust in the Pune monsoon season. Generalising to other crops (wheat, rice), pests (aphids, stem borers), and geographies requires additional data and re-evaluation.
4) Simulation-to-Reality Gap. The sensor noise model is derived from held-out image statistics, not from a real deployed UAV camera system. Real sensors may have correlated noise patterns not captured by our IID noise model.

XIII. CONCLUSION AND FUTURE WORK

This paper presented an autonomous farm spraying system formulated as a Constrained POMDP and solved with Proximal Policy Optimization. The proposed PPO agent was trained and evaluated on a simulator grounded with real UAV soybean imagery (MH-SoyaHealthVision) and real-time Open-Meteo meteorological telemetry. Key findings:
• PPO converges in $N* = 160\,000$ steps and achieves +1.86 reward gain over untreated fields, saving 57 % of chemical compared to blanket spraying.
• DQN achieves higher absolute yield but incurs a negative net reward (-0.60 to -3.84 across scenarios), driven by Q-value overestimation bias toward maximum-dose actions.
• Potential-based reward shaping is critical: removing it reduces the reward gain from +1.62 to +0.10.
• Temporal observation stacking (4-step history) provides robustness to 20 % sensor noise. Future work will explore: (1) continuous dose control with real-valued action spaces; (2) spatial graph neural network encoders for richer field representations; (3) multi-UAV cooperative spraying with shared map observations; and (4) real-world deployment on embedded compute platforms mounted on agricultural rovers.

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APPENDIX A DECISION CYCLE AND INFORMATION FLOW

The report's control loop can be read as a sequence of interfaces rather than as one opaque model. A weather record supplies wind, rainfall, and temperature at the episode time. A route function identifies the current cell. The sensor simulator samples an observation score using the score-bank distribution for the relevant true class. The observation builder combines that local score with neighboring scores, a health estimate, weather indicators, tank proportion, time proportion, and normalized position. The policy receives a stack of recent observations and returns a distribution over the four actions. The selected action is passed through a physical interlock. Only the permitted dose is executed, and the simulator updates the crop, pest, residue, and tank variables before proceeding to the next step.

This sequence provides several points where information can be checked. The route position should agree with the observation's normalized row and column. The weather input should correspond to the intended time index. The selected action should be one of the specified dose values. The interlock should return zero in each blocked condition and should not permit tank exhaustion. The resulting transition should update the same cell and resources that the action targeted. Logging these intermediate values would make it possible to determine whether an unexpected episode result originated in sensing, action selection, masking, or state transition.

The sensor score bank is an abstraction of perception uncertainty. It allows training and evaluation to draw observed scores from a held-out score distribution instead of revealing the latent pest value. In a physical system, the equivalent interface would be a score associated with a specific field cell and acquisition time. If images are processed asynchronously, the controller would need to avoid associating an old score with a new route position. If an image or score is missing, the system should record the absence rather than silently substituting a confident-looking value. These integration details are not described as implemented in the paper, but they follow from the system modules that the report specifies.

The weather stream is also an interface, not merely an additional feature vector. The environment reads wind, rain, and temperature for the current step, while the policy also receives transformed weather features. The interlock should evaluate the units and thresholds before dose execution. In a real deployment, a system would need to identify the source and time stamp of each observation and decide what to do when forecast data disagree with local measurements. The report's hourly Open-Meteo telemetry is the simulation input specified in the study; the paper does not establish the accuracy or latency of a deployed local weather instrument.

The final record for each step could contain the episode identifier, step, cell, raw sensor scores, processed observation, action probabilities, selected action, interlock reason, actual applied volume, weather data, and updated tank level. This record would not make the policy safer by itself, but it would support reproducibility and post-event review. A summary over an episode could then report cumulative return, yield, chemical volume, unnecessary volume, and drift proxy using the same definitions as the benchmark. Such logging would also make differences between commanded dose and measured dose visible if a pump were later integrated.

APPENDIX B INTERPRETATION OF MODEL COMPONENTS

The pest grid represents the hidden disease burden, while the canopy-health grid represents the remaining undamaged crop. Residue represents a temporary protection effect in the simulation, and the tank tracks finite chemical supply. These variables have different roles: the pest process changes over time and across neighboring cells; crop health accumulates damage; residue describes the simulated persistence of treatment; and the tank imposes a shared resource limit. Keeping these variables separate allows the environment to represent a spray action as an intervention that affects both present

disease and future available resources.

The temperature-modulated logistic process contains local growth and neighborhood diffusion. The local term allows pest density to increase as a function of its existing level and temperature response. The neighborhood term allows infestation to influence adjacent cells. The report sets the base growth rate and diffusion coefficient to the stated values and applies a dose-dependent reduction at the treated cell. This is a stylized disease process, not a calibrated epidemiological forecast. The report's limitation section appropriately notes that real fungal spread can be anisotropic and affected by factors absent from the model. The function is useful for controlled policy comparison because all policies encounter the same simulated process, but realism should not be inferred beyond the modeled ingredients.

The observation vector combines local and adjacent disease scores with weather, time, position, and tank state. This design gives the agent information about both immediate need and the context that changes the value of spraying. The score from a neighboring cell can signal a nearby cluster; the tank fraction indicates the remaining resource; time and location indicate how much of the route remains; and weather features indicate whether a dose is currently admissible. Four consecutive observations increase the input size from 13 to 52 and provide recent temporal context. The original report's results and ablations assess that representation within this task; they do not compare every possible recurrent or belief-state architecture.

The reward separates yield change, chemical volume, unnecessary spray, and atmospheric drift. The yield term uses average crop-health change. The chemical-cost term applies to each dose. The unnecessary-spray term applies when the sensor score is below the report's threshold, and the drift estimate scales dose by wind relative to the interlock limit. During learning, potential-based shaping adds a term derived from the potential function over pest density. The report says the shaping term is used only during training and excluded from evaluation. This distinction matters: the evaluation return should represent the unshaped objective, otherwise a policy could appear better partly because of a training aid that is not part of the stated operational objective.

The shaping term is intended to provide a more immediate learning signal when crop-health damage accumulates slowly. In temporal learning, a reward that arrives long after an action can be difficult to associate with that action. A potential difference can assign intermediate feedback based on the change in a state potential while preserving the underlying optimal policy under the conditions described by the cited shaping theory [4]. The report's ablation results show the performance of its tested shaped and unshaped variants. The interpretation remains specific to this implementation; additional reward designs would need their own tests and must not be assumed to preserve the same objective.

APPENDIX C VALIDATION CHECKLIST FOR A FUTURE PHYSICAL TRIAL

A future physical trial should begin by confirming the sensing pipeline on data that are independent at the appropriate level. The evaluation set should be separated by field or acquisition event where possible, and disease labels should be reviewed using a documented procedure. Results should report class counts, error types, and calibration as well as aggregate ranking measures. Because the report uses scores sampled from held-out data, the score-bank construction should be repeatable and should disclose which partitions contributed the score distributions.

The actuator should be validated separately from the learned policy. A test should measure whether each command corresponds to the intended volume under the planned operating configuration. Calibration should be repeated after relevant equipment changes, and the controller should detect conditions where commanded and measured delivery differ. A safe test liquid can be used during initial engineering checks. A field trial should not infer actuator performance from the simulator's subtraction of an action volume from a virtual tank.

The meteorological mask should be tested against boundary values and unit conversions. Tests should include wind immediately below, at, and above the selected threshold, rainfall around its threshold, and missing or stale records. The selected thresholds should be reviewed against product instructions, local regulations, aircraft behavior, and field conditions. The 15 km/h and 1.0 mm/h thresholds are the values in this report's model; they should not be presented as universal recommendations.

The policy should be evaluated in recommendation-only mode before chemical actuation. Experts can compare suggested actions with field observations and identify cases where the state representation is insufficient. During any later supervised trial, an authorized operator should retain a clear stop control and a way to override the action. Every recommendation and override should be logged. Such staged review helps identify mismatches between simulated cell-level decisions and operational realities such as inaccessible zones, sensitive neighboring areas, or crop conditions not represented in the training data.

Finally, the outcome assessment should compare the policy with meaningful alternatives under the same field and weather conditions. The study already uses no-spray, fixed-rate, threshold, context-rule, and Oracle references. For a physical trial, the Oracle remains only a simulator comparator because it observes hidden disease state. A feasible operational comparison should focus on policies that can use the same available inputs, with clear measurement of yield, disease progression, actual pesticide volume, deposition, and off-target movement. Replication across farms and seasons would be needed before making broad generalization claims.

APPENDIX D ADDITIONAL LIMITATIONS OF THE EVIDENCE

The benchmark values summarize a simulator, not a controlled field experiment. Simulation supports repeatable comparison under specified assumptions, but those assumptions determine which effects can appear. The pesticide-cost weight, unnecessary-spray threshold, drift proxy, disease-growth parameters, weather interlocks, sensor-score distribution, and fixed route all shape the learned objective. A different but reasonable setting could alter the balance between yield and chemical use. The reported numerical results should therefore be interpreted together with Table I, the reward definition, and the stated scenario protocol.

The report distinguishes PPO's results across three random seeds from DQN's stated single seed in its hyperparameter table. That distinction affects the strength of conclusions about run-to-run variability. A future comparison should report individual-seed results for every learned policy and use a matched evaluation budget. It should also clarify how evaluation weeks, weather episodes, and pest initializations are sampled. The present report retains its original values and does not claim additional statistical significance beyond its reported summaries.

The dataset and sensor model provide another boundary. A score bank generated from held-out image scores is useful for injecting observed false-alarm and miss behavior into a simulated policy loop. It may not capture all sources of uncertainty in flight, including motion blur, changing illumination, camera calibration, occlusion, cultivar differences, or time-dependent symptom visibility. Likewise, an AUC at cell level or image level does not by itself describe downstream cost when false positives and false negatives have different consequences. Future system-level evaluation should connect those errors to the dose decisions and their agricultural outcomes.

The weather model simplifies both timing and physical response. The report uses hourly meteorological data and binary precipitation and wind interlocks. Local gusts and canopy-level conditions can vary over a shorter time and across the field. Rainfall can also interact with formulation, leaf wetness, and time since application. The simulation represents these factors using the stated thresholds rather than a detailed atmospheric or residue model. No direct drift measurement is reported; the drift quantity is an exposure estimate computed by the reward definition.

The future directions in the original conclusion—continuous actions, graph encoders, multiple UAVs, and embedded hardware—should be considered alongside these evidence limits. They are research directions, not completed capabilities. Expanding the simulator may help assess some design choices, but final claims about environmental effect, crop yield, or safety require field measurements and an appropriately supervised trial. This qualification preserves the value of the current comparative study while avoiding an unsupported leap from model performance to farm-level outcomes.

APPENDIX E OBJECTIVE SENSITIVITY AND DIAGNOSTIC INTERPRETATION

The reward weights define what "better" means within the simulation. The yield coefficient scales the contribution of crop-health change, while the chemical, unnecessary-dose, and drift coefficients price different costs. If a decision-maker changes those weights, the same policy may no longer be preferred. This is not an error in the algorithm; it reflects a different objective. A higher chemical penalty would make chemical conservation more influential relative to crop-health preservation. A higher drift penalty would increase the importance of avoiding actions associated with wind exposure. A higher crop-health value would make disease suppression more valuable. These are directional interpretations of the reward formula, not additional outcomes observed in the reported experiments.

For that reason, a future extension should report a sensitivity surface over plausible reward weights rather than only one selected setting. The same test scenarios could be rerun with a small, predeclared grid of weights, while preserving all other simulator settings. Results could show whether PPO's advantage is stable across choices or concentrated around a narrow setting. Any such experiment should report actual results only after it is run. The present report does not provide that sweep, and this expanded discussion does not imply that it has been performed.

The separation of cost terms also aids diagnosis. If yield is low and chemical use is also low, the system may be too conservative, the score distribution may contain too many misses, or safe weather windows may be scarce. If yield is high but net reward is poor, the extra protection may have required chemical volumes or drift exposure whose penalties outweigh the yield contribution. If unnecessary spray is high, the policy may be acting on false alarms or the sensor threshold may be poorly aligned with the decision costs. If drift exposure is high, the issue may arise from weather-aware behavior, a mismatch in the proxy, or insufficiently frequent local weather updates. These are diagnostic hypotheses to check against logs; they are not unique causal explanations and should not be inferred from a metric alone.

The benchmark should therefore be read as a vector of outcomes rather than a single leaderboard. Table III reports reward, gain, yield, chemical volume, and drift exposure, while the text also discusses unnecessary chemical waste. Each column answers a different question. Reward and gain reflect the specified objective; yield represents crop-health preservation; pesticide volume describes resource consumption; waste relates treatment to the sensor-healthy threshold; and drift is the simulator's estimate. A practical report should retain these measures even when a composite score is used, because a favorable composite could otherwise conceal a behavior important to the operator or regulator.

An additional way to communicate tradeoffs is to show a Pareto frontier. A policy is Pareto-dominated when another policy is at least as good on all selected objectives and

strictly better on at least one. The report's Figure 4 is described as comparing yield and chemical volume, with environmental penalties in a second panel. The "best" point still depends on whether the decision-maker values yield, chemical savings, and environmental exposure equally or applies a specific weighting. The Oracle point is not a feasible target for an operational policy because it has access to the hidden pest state. It is useful as a ceiling within the simulator, but feasible-policy comparisons should remain the basis for deployment decisions.

APPENDIX F STRUCTURED REVIEW OF POTENTIAL FAILURE MODES

An observation failure can occur before the policy chooses an action. A camera may not capture a cell clearly, image mosaicking may associate pixels with the wrong cell, or the disease classifier may return an unreliable score under a condition not represented in training. In the simulation, the score bank represents false alarms and misses through class-conditioned score distributions. A physical implementation would additionally need to detect absent or low-quality images, preserve cell identifiers, and indicate when the score is outside the training support. A safe fallback should avoid treating an invalid score as a confident disease estimate.

A weather-input failure can produce a different class of error. An API record can be delayed, missing, or representative of a location different from the aircraft's current position. The simulation reads weather telemetry by episode time and uses stated thresholds, but does not establish network reliability or field-scale measurement accuracy. A future implementation should attach time and location metadata to every reading, define a maximum acceptable age, and specify what the controller does when it cannot confirm conditions. The operational action should remain blocked when a required safety input is unavailable, unless a separately approved procedure defines another safe mode.

An actuation failure occurs when the applied volume differs from the selected command. A depleted or damaged tank, a blocked nozzle, pressure variation, a pump fault, or a change in vehicle speed could affect delivery. The simulator deducts the specified dose from its virtual tank; it has no pump sensor and does not measure deposition. Consequently, actual volume and actual target delivery should be treated as distinct quantities in future tests. Hardware feedback can be recorded beside the command so that a policy action is not mistakenly credited with a result that the actuator did not execute.

A route or localization failure can direct an otherwise reasonable dose to the wrong cell. The report assumes a fixed serpentine route over the grid. An aircraft in a field would need to track position relative to the map and field boundary, and handle lost localization or an unexpected obstacle. A route error should be logged and should prevent dosing until the target cell is known with adequate confidence. This requirement is not an extension of the learned policy; it is an independent prerequisite for mapping a cell-level action to a physical location.

A policy failure can occur even when sensing, weather, route, and actuation function as expected. The learned action may be undesirable because the test case differs from training, because reward weights encode a poor tradeoff, or because the policy has not encountered a relevant condition. The report's unseen-edge, sensor-noise, and harsh-weather scenarios probe selected sources of distribution shift. They do not exhaust the space of possible failures. Monitoring should therefore retain a no-action or human-review path for observations that lie outside defined operating bounds.

A measurement failure can occur after treatment. Crop damage may progress slowly, residue may not be directly observed, and weather may change after the action. If a report only records the final reward, it can be difficult to identify which mechanism produced it. Step-level logging and repeated field observations can link decisions to later outcomes. At the same time, a monitoring system must avoid claiming that correlation between a dose and subsequent crop state proves that the dose alone caused the outcome. Weather, disease pressure, plant stage, and field heterogeneity can also contribute.

These failure categories suggest a layered review: validate the input data, validate the decision logic, verify the interlocks, measure the actuator, verify the target location, and assess post-treatment outcomes. This layered review is consistent with the report's modular design and with aerial application guidance that treats planning, equipment, weather, training, and record keeping as parts of one operation [17]. Drift research emphasizes that environmental movement has multiple interacting determinants, so a simulator proxy should be checked against measured deposition and local conditions before it is used as a field assurance [18].

APPENDIX G REPORTING CHECKLIST FOR EXTENDED EXPERIMENTS

An expanded experiment should identify the training and evaluation populations. It should state which infestation patterns are available during training, which are reserved for testing, how weather weeks are assigned, and whether multiple images from one field or flight can appear in different partitions. It should report the number of episodes per condition and how returns and agricultural metrics are aggregated. These details determine what "unseen" means and how much evidence supports a robustness statement.

The experiment should report all algorithm settings that influence the update process, including learning rates, batch or rollout size, number of update epochs, discounting, exploration, target-network updates, and random seeds. The paper provides many of these values in Table II. A replication package should also document library versions, network initialization, observation normalization, action-mask implementation, and checkpoint selection. If the best checkpoint is selected using a test scenario, that selection can make the test result optimistic; model selection should use training or

validation data and reserve the reported test conditions for final evaluation.

For performance reporting, each policy should be assessed on the same evaluation episodes where feasible, so differences are not confounded by a different set of initial infestations or weather sequences. Results should include seed-level variability as well as means. A confidence interval or distribution plot can complement the mean, especially when rare adverse conditions are important. The current paper reports PPO bands across three seeds for its learning curve and states one DQN seed in the hyperparameter table; an extended study could use matched seed counts and publish the individual runs. This recommendation does not revise or reinterpret the existing table values.

The physical metrics should have explicit units and computation rules. Yield preservation should identify its denominator and whether it is an episode-level or field-level fraction. Chemical use should distinguish planned volume, volume commanded, and volume measured. Unnecessary dose should state the score threshold and whether the classification label or noisy score determines the penalty. Drift exposure should state the proxy and should not be confused with directly measured off-target mass. The present report supplies definitions for its simulated metrics, and a field study would need to add measurement protocols for physical outcomes.

Finally, an extended paper should report failed runs and implementation limitations rather than only successful episodes. If a policy collapses, exhausts its budget, or repeatedly triggers an interlock, that behavior is part of the evidence. A safety-sensitive system should also report the rate and reason for rejected actions. Such reporting makes it possible to distinguish a policy that performs well because it learned to respect constraints from one that only appears successful because the environment silently removes unsafe actions. The report describes its mask in the environment and includes the interlock in the training pseudocode; a future implementation should additionally quantify how often each mask condition is activated.

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